

Growth Driver Deep Dive: Aerospace, Space and Defence

Reframed with quantified government targets and a named company case study: Fluidyne's documented HAL/ISRO/BARC track record

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry,
leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. Upgrades this driver from a generic mention to a primary growth thesis with a live proof point.

16 — Reframed Growth Driver: Aerospace, Space & Defence Indigenisation

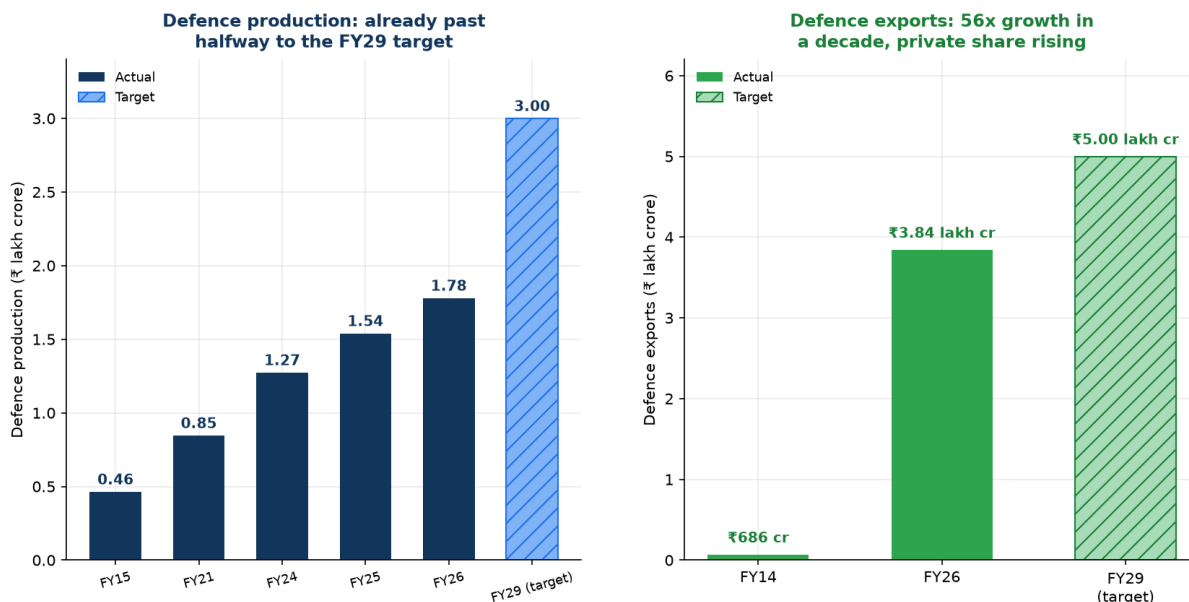
In [12-india-drivers-challenges.md](#), this was driver #9 out of 10 — one line, "Medium" fit, ranked below steel and nuclear. That ranking undersold it. On closer research it deserves to sit near the top, for a reason the earlier version missed entirely: **this is the one driver where an Indian bellows maker has an already-documented, named track record of supplying a flagship national programme.** Every other driver in this pack is inferred from policy. This one has a receipt.

The one-paragraph reframe

India's defence production has grown 15.6% year-on-year to ₹1.78 lakh crore in FY26 — already past the halfway mark of a ₹3 lakh crore target for FY29 — while defence exports have grown 56x in a decade to ₹38,424 crore. Space gets a steadily rising ₹13,706 crore budget now shifting into hardware realisation for Gaganyaan and a new heavy-lift launch vehicle. And the indigenous fighter engine programme — India's most demanding metallurgical qualification challenge outside semiconductors — has just opened an eleven-year supplier-development window as private industry is brought into the AMCA programme for the first time. All three verticals demand exactly the property set Iwatani already trades: thin-gauge, high-fatigue, corrosion-resistant stainless and nickel-alloy strip. And unlike every other driver in this pack, this one is not hypothetical: **Fluidyne Engineers of Mysuru is a registered vendor to ISRO, HAL, BARC, DRDO, NPCIL, BEL, BHEL and the Navy, and has been publicly honoured by HAL Nasik for supplying the LCA Tejas Mk1A programme.**

1. Defence: government targets, already half-achieved

Chart 32 — India's defence industrial base: government targets, not analyst forecasts



[P/C] Ministry of Defence / PIB / Rajnath Singh statements, FY2025-26 review. Production: ₹46,429 cr (FY15) -> ₹84,643 cr (FY21) -> ₹1,27,434 cr (FY24) -> ₹1.54 lakh cr (FY25) -> ₹1.78 lakh cr (FY26, +15.6% y/y) -> ₹3 lakh cr target (FY29). Private sector share of production reached an all-time high of 24% (₹42,000 cr) in FY26, up from 22% in FY25, and has nearly tripled since FY17. Exports: ₹686 cr (FY14) -> ₹38,424 cr (FY26, +63% y/y) -> ₹50,000 cr target (FY29).

Metric	FY2015	FY2024	FY2026	FY2029 target
Defence production	₹46,429 cr	₹1,27,434 cr	₹1.78 lakh cr (+15.6% y/y)	₹3 lakh cr
Defence exports	₹686 cr (FY14)	—	₹38,424 cr (+63% y/y)	₹50,000 cr

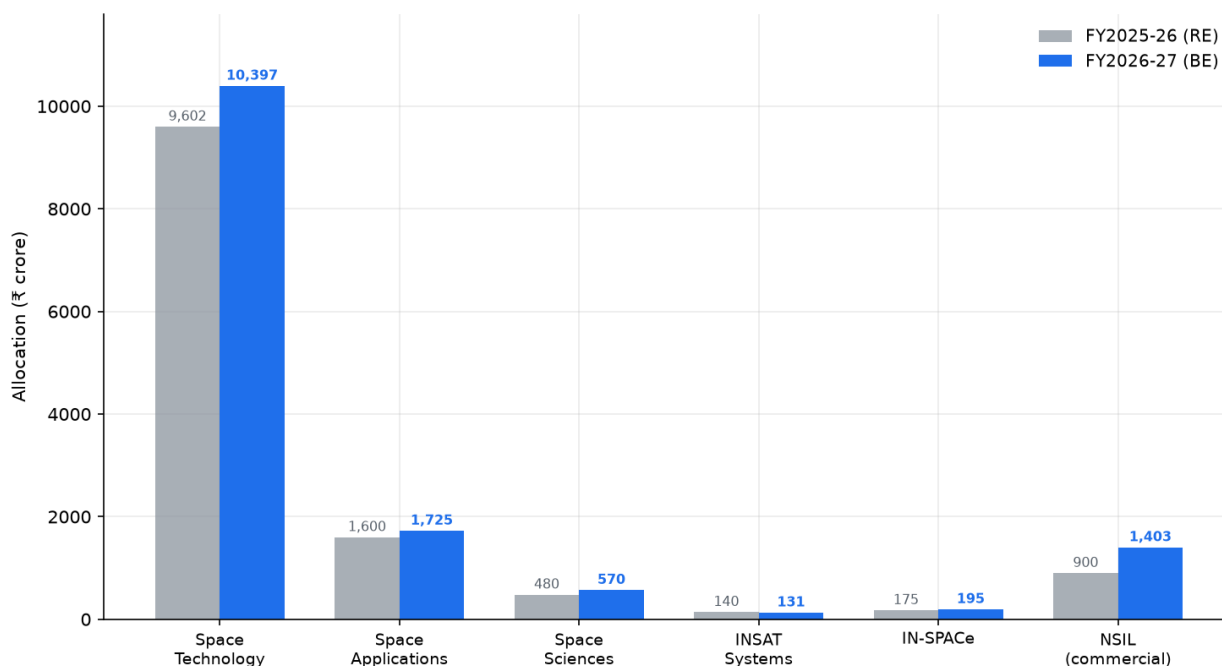
Sources: [Ministry of Defence FY26 review via Rediff \[C\]](#); [PIB — Make in India Powers Defence Growth \[P\]](#); [The Policy Edge — 12-year MoD review \[C\]](#); [Rajnath Singh statement via SSBCrack \[C\]](#).

Three details that matter more than the headline number:

- **Production has already crossed the halfway mark to the FY29 target**, two-plus years early. This is not an aspirational plan; it is a programme ahead of schedule.
- **The private sector's share of defence production reached an all-time high of 24% (~₹42,000 crore) in FY26**, up from 22% in FY25, and **private-sector output has nearly tripled since FY17** while total domestic production is up 140% over the same period. The growth is disproportionately private, which is exactly the part of the supply chain a trading house can reach — DPSUs run closed internal supply chains; private primes and Tier-1s build open vendor bases.
- **Positive Indigenisation Lists** now cover **5,012 items** across five lists from Defence PSUs plus 509 items across five lists from the Armed Forces directly, with **more than 3,000 items already indigenised** and a sixth list "soon to be notified." A Positive Indigenisation List is a government notification that a category of component **must be sourced domestically after a stated date** — this is the regulatory mechanism that converts "aerospace demand" from a vague driver into named, dated, mandatory local-sourcing categories. **Finding out which PIL categories include bellows-adjacent items (seals, expansion joints, flexible couplings, actuator components) is a concrete, high-value research action.**

2. Space: modest budget, but the capital mix is turning to hardware

Chart 33 — India's space budget: ₹13,705.63 crore for 2026-27, capital spend up as Gaganyaan and NGLV move to hardware realisation



[P] Union Budget 2026-27, Department of Space. Total outlay ₹13,705.63 crore (+2% over FY26 BE), of which capital expenditure ₹6,375.92 crore (+20% over FY26 RE) — signalling the shift from preparatory work to hardware for Gaganyaan, the Next Generation Launch Vehicle, and planetary missions (Chandrayaan-4, LUPEX, Venus Orbiter). NSIL budgetary support rises to ₹1,403 crore to drive commercial revenue through private-sector production and technology transfer.

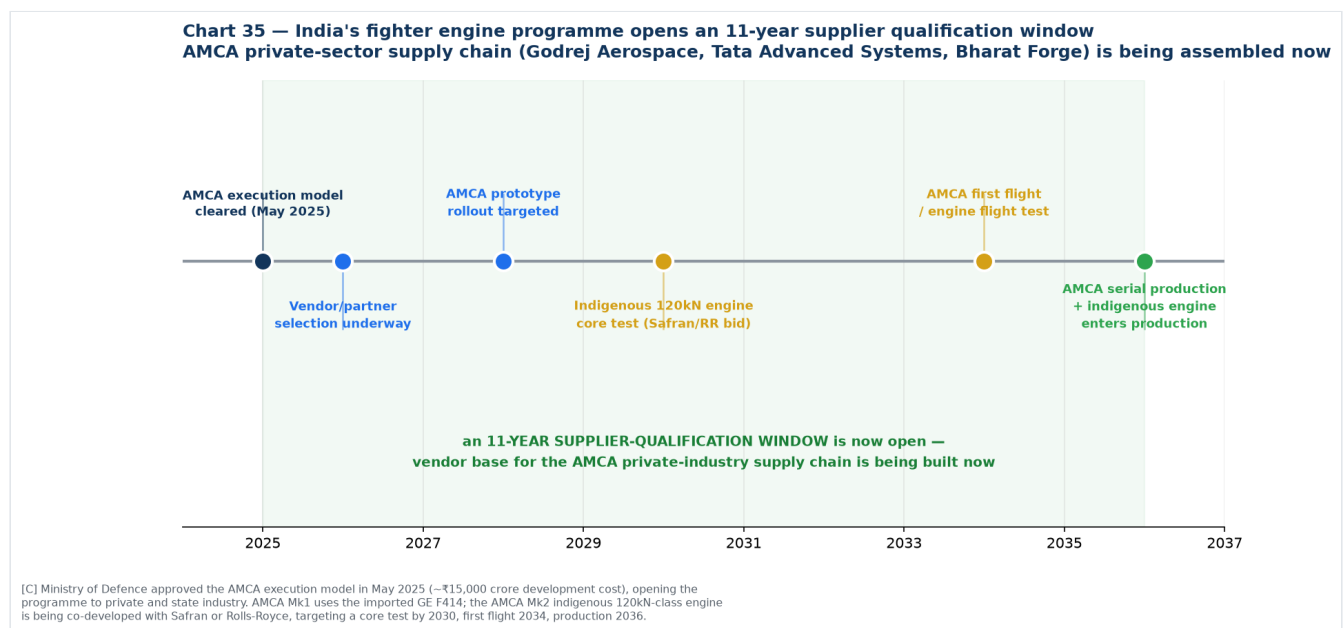
The Department of Space received **₹13,705.63 crore for 2026-27**, a modest 2% increase in total, but with **capital expenditure up 20% to ₹6,375.92 crore** ([Union Budget 2026-27 via The Hindu](#), [CNBC-TV18](#), [Times of India \[P/C\]](#)).

Why the capex shift matters more than the topline: officials describe this explicitly as the programme moving "from preparatory work to hardware realisation" for the **Gaganyaan** human spaceflight programme, the **Next Generation Launch Vehicle (NGLV)**, and planetary missions (**Chandrayaan-4**, **LUPEX**, **Venus Orbiter**). Hardware realisation means fabrication contracts — cryogenic engine components, propellant feed lines, actuator bellows, and gimbal assemblies — moving from design to procurement.

NSIL (ISRO's commercial arm) support rises to ₹1,403 crore, explicitly to "drive commercial revenue through private-sector production and technology transfer." NSIL has already demonstrated the anchor-customer model: it is paying **₹3,000 crore** for the GSAT-7B communication satellite and using the **HAL-L&T consortium** to build PSLVs on a public-private-partnership basis ([Anatomy of India's 2026-27 Space Budget \[C\]](#)). That is the structural sign of a maturing procurement model — government demand that private industry can build a balance sheet against.

Private space is also live independently of the budget. Agnikul Cosmos achieved India's first successful hot-fire test of a clustered configuration of three 3D-printed semi-cryogenic engines ([search summary context](#)) **[C]** — and cryogenic/semi-cryogenic engines are a canonical bellows application (propellant feed line flex joints, gimbal actuator bellows), so a growing set of private Indian launch vehicle companies is now a second potential customer category alongside ISRO itself.

3. The fighter engine programme: an 11-year supplier-qualification window just opened



This is the single most technically demanding — and most overlooked — item in the whole India driver set, because engine-grade bellows and flexible couplings sit at the extreme end of the material specification this entire pack is built around.

The programme in outline:

- **May 2025:** the Ministry of Defence cleared the **AMCA (Advanced Medium Combat Aircraft) execution model**, opening India's fifth-generation stealth fighter to private industry for the first time — HAL is not leading it. Development cost is estimated at **~₹15,000 crore** ([19FortyFive \[C\]](#)).
- **Private industry partners being positioned** under the Development-cum-Production Partner framework include **Godrej Aerospace, Tata Advanced Systems, and Bharat Forge** ([search synthesis \[C\]](#)).

- **AMCA Mk1** will fly on the imported GE F414 (a fresh **\$1 billion, 113-engine deal** with GE was signed in November 2025, delivering 2027–2032) ([19FortyFive \[C\]](#)).
- **AMCA Mk2** requires a new **110–120 kN-class indigenous engine**, being co-developed with either **Safran** (France, full technology transfer under discussion) or **Rolls-Royce** (final offer submitted: engine core test by 2030, first flight 2034, production 2036, under an **80% technology-transfer** structure similar to the existing GE-HAL F414 arrangement) ([Rolls-Royce pitch](#), [ThePrint \[C\]](#); [DRDO/GTRE timeline \[C\]](#)).

Why this is directly relevant to bellows. A jet engine's fuel and hydraulic systems, actuator feedback linkages, bleed-air ducting and afterburner nozzle actuation all use precision metal bellows and flexible couplings, typically in Inconel or precipitation-hardening stainless for high-temperature fatigue resistance — the exact material class covered in [01-technical-primer.md](#). Whichever OEM wins the AMCA Mk2 engine, a **technology-transfer structure with an Indian manufacturing partner is now the explicit government requirement**, which means an Indian supply chain for engine-grade components is being built from a standing start, on a clear multi-year timeline.

The commercially useful fact is the timeline itself. Core test 2030, first flight 2034, production 2036 is not a near-term order book — it is an **eleven-year window in which the private-industry vendor base for this programme is being assembled**. Qualification cycles for flight-critical components run for years; the vendors who begin qualification conversations now will be positioned when volume production starts. This is a "get in line early" opportunity, not a "there is an order today" one.

4. The proof point: Fluidyne is not a hypothetical customer

Chart 34 — This is not hypothetical: an Indian bellows maker already supplies India's fighter jet programme

Fluidyne Engineers (India) Pvt Ltd, Mysuru — registrations, recognitions and named programme deliveries, from the company's own site

REGISTERED VENDOR TO

- DGQA — Directorate General of Quality Assurance
- ISRO — Indian Space Research Organisation
- HAL — Hindustan Aeronautics Limited
- BARC — Bhabha Atomic Research Centre
- NPCIL — Nuclear Power Corporation of India
- BEL — Bharat Electronics Limited
- BHEL — Bharat Heavy Electricals Limited
- MDL — Mazagon Dock Shipbuilders Limited
- IRS — Indian Register of Shipping
- EIL, PDIL, ILP, Indian Coast Guard, Ship Building Centre

NAMED PROGRAMME DELIVERIES

HAL Nasik honour

Precision pipe bending for LCA Mk1A (Tejas) and HTT-40 trainer aircraft

SIATI award

Indigenous development of metallic bellows for frontline trainer aircraft

IDEMI recognition

Precision welding, Inconel to SS304, for BARC

Materials qualified

Titanium bellows supplied to valve manufacturers and atomic energy research organisations

Industry ranking

Top 10 Expansion Joint Manufacturers, Industry Outlook

Source: [fluidyneengineers.com/certification.php](#), [fluidyneengineers.com/about.php](#) — company's own published record [P]

Every other driver in this file describes a policy tailwind and infers that it should help Indian bellows makers. This one does not need to infer anything, because **Fluidyne Engineers (India) Pvt Ltd, Mysuru** — already identified in this pack as a Track A customer (see [07](#) and [10](#)) — publishes its own strategic-sector record:

Registered vendor to: DGQA, **ISRO**, **HAL**, **BARC**, **NPCIL**, BEL, BHEL, MDL, IRS, EIL, PDIL, ILP, Indian Coast Guard, Ship Building Centre Visakhapatnam ([Fluidyne certifications page \[P\]](#)).

Named, dated programme deliveries:

Recognition	What it actually was
Honoured by HAL Nasik	Precision pipe bending for LCA Mk1A (Tejas) and HTT-40 trainer aircraft — HAL's own flagship fighter and trainer programmes
SIATI award	For indigenous development of metallic bellows for frontline trainer aircraft — a direct, named bellows application in a military aircraft programme
IDEMI recognition	Precision welding of Inconel to SS304 for BARC — nuclear-grade dissimilar-metal joining
Materials qualification	Titanium bellows supplied to valve manufacturers and atomic energy research organisations

Source: [Fluidyne certifications and recognitions \[P\]](#) — the company's own published record. This has not been independently verified beyond the company's self-disclosure, and as with other Fluidyne claims in this pack, a site visit would confirm scale and current activity. But it is a materially stronger evidence base than a policy inference: **it names a specific aircraft type, a specific plant (HAL Nasik), and a specific award citation for bellows specifically.**

Why this matters for the strategy. This is direct evidence that the qualification pathway from "Indian precision bellows maker" to "flagship aerospace programme supplier" **has already been walked once**, by a company already on Iwatani's target customer list for entirely separate reasons (UHV, 0.05mm diaphragms — see [04](#)). If Fluidyne's aerospace bellows business is growing — and Fluidyne's overall revenue grew **+37% FY24→FY25** (see [10-india-market-data.md](#) §6) — then supplying its material inputs is a doubly-motivated customer relationship: it serves both the rung-1 mechanical-seal/industrial thesis and this aerospace/defence thread simultaneously.

5. What this means for the driver ranking

	Original treatment (file 12)	Reframed
Position in ranking	#9 of 10, "Medium" fit	Should move to top 4 , alongside refining, hydrogen and Semicon 2.0
Evidence type	Generic reference to indigenisation lists and BARC/ Navy/DRDO customer mentions	Quantified government targets (₹3 lakh cr production, ₹50,000 cr exports, both already 50%+ achieved) + a named, dated company case study
Iwatani relevance	Stated as "qualification-heavy"	Correct, but now specific: the AMCA engine timeline (2026–2036) is a defined qualification window, not a vague future opportunity
Actionability	None specified	Concrete: (1) identify which Positive Indigenisation List categories cover bellows/seals/couplings, (2) approach Fluidyne specifically citing its own HAL/ BARC/ISRO record, (3) track AMCA Mk2 engine partner announcement for the Indian Tier-1 name to approach

6. Three actions this reframing generates

- 1. Pull the actual Positive Indigenisation List text** (available from the Ministry of Defence / Department of Defence Production) and search for aerospace bellows-adjacent categories: seals, gaskets, expansion joints, flexible couplings, actuator sub-assemblies, hydraulic/pneumatic line components. This converts "defence indigenisation is a driver" into a named list of mandatory-local-source part categories — the single most concrete piece of evidence this pack could still add.
- 2. Approach Fluidyne with its own case study**, not a generic pitch. A conversation that opens with "we understand you supplied bellows for the HTT-40 and Tejas Mk1A programmes recognised by HAL Nasik and SIATI" is a materially stronger opening than a cold material-supply pitch, because it demonstrates Iwatani has done the work to understand the customer's actual, differentiated business.

3. **Track the AMCA Mk2 engine partner decision** (Safran vs Rolls-Royce, expected before end-2026 per the Rolls-Royce final-offer timeline). Whichever OEM wins will name an Indian manufacturing and technology-transfer partner, and that partner will need to build an engine-grade component supply chain from very close to zero — this is the single highest-value name to identify and approach early in the whole India driver set.
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New sources introduced in this file

Source	URL	Grade
PIB — Make in India Powers Defence Growth	https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/apr/doc202543531401.pdf	[P]
Rediff — India's defence output hits new high (FY26 review)	https://www.rediff.com/business/report/indias-defence-output-hits-new-high/20260618.htm	[C]
The Policy Edge — 12-year MoD review 2014-2026	https://www.policyedge.in/p/how-indias-defence-sector-changed-between-2014-and-2026	[C]
SSBCKrack — Rajnath Singh, ₹3 lakh crore target, PILs	https://www.ssbckrack.com/2026/07/defence-production-target-of-rs-3-lakh-crore-by-2029-rajnath-singh-highlights-indias-defence-transformation-after-operation-sindoor.html	[C]
Fortune India — defence Aatmanirbhar push	https://www.fortuneindia.com/economy/defence-aatmanirbhar-push-centre-plans-indigenous-production-worth-3-lakh-cr-50000-crore-exports-by-2029/128330	[C]
The Hindu — Union Budget 2026, space programme	https://www.thehindu.com/sci-tech/science/union-budget-space-programme-budget-allocation-isro/article70577352.ece	[P/C]
CNBC-TV18 — India's space budget 2026-27	https://www.cnbctv18.com/budget/indias-space-budget-inches-up-in-2026-27-capital-spend-rises-but-ambition-outpaces-funding-19838009.htm	[C]
Times of India — Budget 2026 space allocation	https://timesofindia.indiatimes.com/business/india-business/budget-2026-space-department-allocation-rises-to-rs-13706-crore-capital-spend-gains-pace/articleshow/127838489.cms	[C]
SatNews — India private space budget triggers	https://satnews.com/2026/02/01/indias-private-space-industry-sector-finds-indirect-growth-triggers-in-union-budget-2026-27/	[C]
Anatomy of India's 2026-27 Space Budget (NSIL, anchor-customer model)	https://orrery.ashwinprasadrao.com/p/anatomy-of-indias-2026-27-space-budget	[C]
The Diplomat — AMCA engine ecosystem	https://thediplomat.com/2026/06/why-the-heart-of-indias-indigenous-fighter-ecosystem-will-be-american-for-several-decades-to-come/	[C]
ORF — IAF fighter engine conundrum	https://www.orfonline.org/research/addressing-the-iaf-s-fighter-engine-conundrum	[C]
19FortyFive — AMCA programme, F404 deal, execution model	https://www.19fortyfive.com/2026/06/india-has-30-brand-new-fighters-it-cant-fly-and-its-promising-a-stealth-jet-by-2035/	[C]
ThePrint — Rolls-Royce AMCA engine pitch	https://theprint.in/defence/engine-core-by-2030-test-flight-by-2034-production-by-2036-rolls-royce-makes-final-pitch-to-power-amca/2967532/	[C]
The Defense News — DRDO/ GTRE AMCA Mk2 engine timeline	https://www.thedefensenews.com/DRDO-Says-AMCA-Mk-2-Engine-Could-Be-Ready-by-203536-If-CCS-Clears-Project-This-Year/	[C]
Fluidyne Engineers — certifications and recognitions (HAL, SIATI, IDEMI)	https://www.fluidyneengineers.com/certification.php	[P] — company's own published record
Fluidyne Engineers — about (aerospace facility, 100+ components delivered)	https://www.fluidyneengineers.com/about.php	[P]
Fluidyne Engineers India — profile (industries served)	https://www.fluidyneengineersindia.com/profile.html	[P]
Industry Outlook — Fluidyne interview (materials, BARC/ Navy/DRDO supply)	https://www.theindustryoutlook.com/manufacturing/vendor/fluidyne-engineers-offering-qualitydriven-custom-built-precision-metal-bellows-cid-2238.html	[C]

Chart index

Chart	File	Purpose
32	32-defence-production-exports.png	Defence production and export trajectory, actual vs FY29 target
33	33-space-budget.png	Space budget by head, FY26 vs FY27, capex shift to hardware
34	34-fluidyne-proof-point.png	Fluidyne's documented strategic-sector vendor status and named deliveries
35	35-amca-timeline.png	AMCA engine programme timeline and the 11-year supplier window

Generated by [charts/make_defence_charts.py](#).